FPP Report

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1. "Own work" statement

The program submitted by me for the CS-1CT Free Programming Project is my own work, apart from the following parts, which I explain in the *How it works* section below:

1. I used the tkinter to import images in GIF file and (screen.cv.bind) to control mouse movement which is not covered in lectures
2. For the Turtle interactions, I followed the official Python documentation for things like onkey(), ontimer() and mouse tracking through the canvas
3. The file handling parts such as opening and appending to ingredients.txt and leaderboard.txt follow standard Python practices I learnt both from lab and lectures but all the implementations were written by me.

2. What? Chef's Playground: Interactive Plating Game

Che's Playground is an interactive cooking themed game build using python's turtle graphic system and multiple of standard libraries. The player is given a randomly generated "order" and must select the correct ingredients. Then, the player needs to drag the ingredient onto a drawn plate using the mouse and place them with their own creativity before the confirmation.

The game includes several features:

- A plate rendering interface by turtle and Ingredients loaded from a text file provided
- Ingredient graphics is movable following the mouse cursor and placement confirmation mimicking real game
- Order generation and scoring with up-to date leaderboard system
- Additional feature to add new ingredients to make the game more fun

3. What motivated you to choose this project

I chose this project mainly because I wanted to create something that's actually fun to look and interactive instead of just another text-only program. I've learnt a lot of python basics in the lectures with some additional self-learnt things like mouse tracking, input the image file and sort function for the leaderboard and I felt like a good opportunity to put them together. On my personal side, I've been recently learning cooking and plating for the past few months because I hope to open a small café or restaurant in the future as side income. Thus, creating a food themed project is meaningful for me and it's not just something random comes from my mind. Throughout the project progression, I realised it's really cool if one day I could make a small interactive game for people who visit my café. Finally, this project is a good way for me to build and practice thinking critically for my programming skills and keep moving towards to my long-term goal of becoming software engineer.

4. Brief user instructions

- First of all, ensure all the files (ingredients.txt, leaderboard.txt, FinalProject.py and other .gif files) must be in the same folder and directory. Open the terminal and run the game. Then, you'll see main menu (1. Play game 2. Add ingredients and 3. Exit).
- If you choose **1**, the first thing you need to do is to **split the IDLE and turtle screen into two ¾ for IDLE/ Visual code and ¼ for turtle.**

- Read the rules and follow the terminal carefully, make sure you click on turtle screen before placement confirmation for the ingredients.
- You can use "PizzaA.gif" to test the ingredient adding function and "Corriander.txt" and "BoiledEgg.txt" to add new items in the available category. You can find any transparent or crop images and convert them into GIF format before adding them in the program folder.

5. How it works

Data structures

1. **Dictionary of dictionaries**

The outer key represents the ingredients categories and inner keys represent the specific ingredient names. The value is a function that loads the corresponding image which can be used by turtle for displaying the ingredient. This allows quick look-up for ingredients by category and name and make it easy to load images.

2. **List**

order is a list of ingredients randomly selected for each game using **random.sample()**. The list is used to compare the player's ingredients selection with the required order to calculate the score.

3. **Strings**

Strings was used for storing file paths, player names and others input and output such as **"chicken.gif"** and Playername:Score% from "leaderboard.txt"

4. **Leaderboard dictionary (scores)**

The key is the player's name and the value (Highest recorded score). This dictionary is to keep tracks of player performance across multiple game sessions and it's sorted to display ranked leaderboard.

Interesting / tricky / subtle aspects of the code meriting further explanation

1. **Plate Boundary**

The game uses mouse tracking to make ingredients follow the cursor smoothly. Each ingredient is represented by a Turtle object that constantly move to the direction of the mouse in small steps. A timer ensures continuous movement without freezing the program. Ingredients will stop when the player press "c" on keyboard. I ensure ingredients are prevented from leaving the plate to make the game seems more realistic. The program checks their distance from the plate's centre and redirects the cursor if they move too far.

2. **File handling**

The ingredients file stores each ingredient's category's name and image and there's function to add new ingredient during the gameplay. The leaderboard file tracks player names and scores. The file will be updated if player achieve a higher score and there's a sorting function to show the top player first.

3. **Score calculation**

The game will generate random order of ingredients ranging from 2 to 5 without repetition. Ingredient's images are loaded one by one smoothly only after the user input their choice. Scores are then calculated based on how many ordered ingredients the player used.

6. Particular challenges I overcame

At first, I planned to draw everything from scratch but to my disappointment, moving drawn object is impossible after I finished 3 turtle drawings. I decided to learn everything on Youtube and some others python documentation to make my project look different than the usual I write during labs. The most challenging parts from this project was making the ingredients move smoothly with the mouse without lag and ensure the image only stay on the plate not anywhere else. Furthermore, managing the leaderboard was really tricky because I had to ensure the scores updated correctly and only the best score for each player was updated.